

Proposed U.S. Job Description

Capital Group CDM Modernization Project

L-1B Specialized Knowledge Assignment | LTIMindtree

Employee	Nikhil Subhash Shinde
Employer	LTIMindtree
Client / Engagement	Capital Group - CDM Modernization
Proposed U.S. Position	Senior Data Engineer / Technical Lead - CDM Cloud Data Modernization Specialist
Primary Specialized Knowledge Theme	LTIM proprietary platform knowledge + LTIM modernization methods + Capital Group CDM project-specific implementation knowledge
Prepared For	LTIM Immigration Team / External Immigration Counsel

Confidentiality Note: This document is intentionally drafted at a business and technical responsibility level. It avoids disclosure of confidential Capital Group data, detailed schemas, production identifiers, client credentials, or non-public implementation artifacts. Bracketed placeholders should be completed or removed by LTIM immigration counsel before final submission.

1. Position Overview

The proposed U.S. role is a specialized-knowledge technical leadership position supporting the Capital Group CDM Modernization Program. The assignment requires continuity of LTIMindtree-specific modernization practices, knowledge of LTIM proprietary platforms such as Canvas AIOps and Canvas Insights, and project-specific implementation knowledge acquired through direct participation in the CDM modernization work.

The role is not intended to fill a generic Senior Data Engineer position. The assignment requires an employee who already understands LTIM's modernization delivery model, the CDM conversion context, Informatica-to-Python/PySpark transition patterns, workflow dependency behavior, validation controls, deployment readiness needs, and the specialized accelerators applied to reduce migration risk.

Proposed Position Summary

As Senior Data Engineer / Technical Lead - CDM Cloud Data Modernization Specialist, Nikhil Subhash Shinde will provide technical leadership for the modernization of legacy Informatica PowerCenter and AutoSys-driven ETL workflows into cloud-native Python/PySpark, Databricks, AWS, and Airflow-aligned data pipelines. The role will support architecture analysis, dependency assessment, reusable framework design, code conversion guidance, reconciliation, validation, operational readiness, and knowledge transfer between the offshore and U.S. delivery teams.

2. Project Background

Project	Capital Group - CDM ETL / Data Modernization
Business Domain	Fixed Income, Equity Data, CDM modernization, enterprise data engineering, lakehouse-aligned architecture
Legacy Environment	Informatica PowerCenter workflows, sessions, mappings, transformations, source-target definitions, dependency structures, and AutoSys-based scheduling/orchestration
Modernization Target	Python/PySpark-based reusable pipelines, Databricks/AWS-aligned execution patterns, Airflow orchestration, CI/CD deployment, reconciliation controls, audit tracking, and production-readiness practices
Primary Objective	Preserve existing business logic while converting legacy ETL execution into scalable, maintainable, cloud-native modernization patterns
U.S. Assignment Need	Ensure delivery continuity, reduce ramp-up risk, support onsite/offshore coordination, and apply already-acquired CDM implementation knowledge to client-facing modernization execution

3. Primary Job Duties and Time Allocation

Time	Duty Area	Detailed Responsibility
20%	CDM Modernization Technical Leadership	Lead technical analysis and solutioning for the Capital Group CDM modernization workstream. Guide conversion of legacy Informatica workflows into Python/PySpark and cloud-native execution patterns while preserving business logic, data controls, dependency behavior, and operational intent.
15%	Informatica Workflow and Dependency Analysis	Analyze Informatica PowerCenter XML assets, workflows, sessions, mappings, sources, targets, transformations, and dependency metadata. Identify conversion rules, downstream impacts, reusable patterns, and risk areas that must be addressed during modernization.
15%	Python/PySpark Framework and Conversion Design	Design and guide reusable Python/PySpark components for ingestion, transformation, configuration management, validation, logging, audit tracking, exception handling, and publishing. Standardize conversion output so that multiple workflow groups can follow a consistent engineering model.
15%	LTIM Canvas AIOps and Canvas Insights Alignment	Apply LTIM proprietary platform knowledge to support production-readiness and SDLC traceability. Use Canvas AIOps concepts for alert correlation, incident intelligence, operational governance, digital SOP thinking, and proactive monitoring. Use Canvas Insights concepts for requirement-to-code traceability, workflow impact analysis, testing impact, and delivery visibility.
10%	Reconciliation, Validation, and Data Quality Controls	Define and apply source-to-target reconciliation and validation practices between legacy Informatica outputs and modernized Python/PySpark/cloud outputs. Support evidence-driven validation, data quality checks, exception analysis, audit controls, and equivalence proof for migrated flows.
10%	Orchestration, Deployment, and Production Readiness	Support Airflow/Databricks orchestration patterns, CI/CD deployment readiness, runbook alignment, monitoring readiness, failure handling, logging standards, and post-deployment operational support practices for modernized pipelines.
10%	Onsite/Offshore Coordination and Knowledge Transfer	Act as a technical bridge between the U.S. onsite stakeholders, LTIM program leadership, offshore engineering teams, architects, data analysts, QA teams, and production support teams. Clarify transformation behavior, conversion decisions, dependency sequencing, and release expectations.
5%	Documentation, Governance, and Risk Mitigation	Prepare and maintain technical notes, migration decision records, validation summaries, dependency documentation, design walkthroughs, evidence packs, and governance materials needed for controlled modernization delivery.

4. Detailed Responsibilities

Responsibility Area	Description
Modernization assessment	Conduct detailed review of Informatica PowerCenter workflows, sessions, mappings, transformations, source-target definitions, parameter files, job dependencies, and historical execution patterns.
Conversion strategy	Translate legacy Informatica design patterns into Python/PySpark-aligned processing logic while preserving transformation semantics, load sequence behavior, data validation rules, and exception handling requirements.
Reusable framework design	Guide development of reusable data engineering modules for ingestion, standardization, transformation, publish-layer generation, reconciliation, audit logging, configuration management, and operational controls.
Canvas AIOps operational alignment	Apply Canvas AIOps knowledge to strengthen production-readiness through alert convergence thinking, CI/service impact awareness, issue category identification, digital SOP design, workflow automation readiness, and proactive operational monitoring.
Canvas Insights delivery alignment	Apply Canvas Insights concepts to support SDLC knowledge fabric, requirement-to-code traceability, workflow dependency analysis, impact assessment, defect/test linkage, test optimization, and delivery transparency.
Reconciliation and validation	Use project-specific validation approaches and accelerators to compare legacy and modernized outputs, identify mismatches, support root-cause analysis, and produce evidence that migrated pipelines preserve expected results.
Orchestration migration	Support movement from legacy scheduling concepts to Airflow/Databricks-aligned orchestration, including dependency modeling, trigger design, restart behavior, failure handling, and SLA visibility.
Deployment readiness	Collaborate on CI/CD and release governance practices, including package promotion, environment readiness, deployment notes, monitoring checkpoints, runbooks, and operational handoff.
Stakeholder support	Participate in technical discussions with onsite and offshore stakeholders to clarify business logic, resolve ambiguous mapping behavior, assess change impacts, and drive decisions required for modernization execution.
Knowledge continuity	Transfer accumulated LTIM and CDM implementation knowledge to U.S. and offshore teams so the modernization program can continue without avoidable ramp-up delay or repeated discovery effort.

5. Specialized Knowledge Required for the Position

The proposed U.S. assignment depends on three integrated knowledge areas. The percentages below describe the relative basis of the role's specialized knowledge, not a strict daily time allocation.

Weight	Specialized Knowledge Area	Role Relevance
40%	LTIMindtree Proprietary Platforms & SDLC Methodologies	Canvas AIOps, Canvas Insights, LTIM modernization delivery approach, legacy-to-cloud conversion methods, SDLC traceability, operational readiness, incident intelligence, governance, and production-support thinking specific to LTIM's delivery model.
35%	Capital Group CDM Project-Specific Implementation Context	Confidential understanding of the CDM modernization architecture, workflow dependencies, schema behavior, transformation logic, orchestration strategy, deployment model, validation controls, and production-readiness expectations acquired through direct project execution.
25%	Project-Specific Engineering Accelerators & Validation Frameworks	FastRecon, informatica-python, informatica-sparker, and reusable Python/PySpark framework patterns used as supporting evidence of advanced engineering capability and modernization contribution. These accelerators support reconciliation, Informatica metadata extraction, conversion standardization, and validation quality, but they are not presented as the sole basis for L-1B specialized knowledge.

Specialized Knowledge Positioning Statement

Nikhil's proposed role is strongest when framed around integrated transfer of LTIM-specific modernization know-how. His knowledge combines LTIM proprietary platform capability, Capital Group CDM implementation experience, and project-specific engineering accelerators that package modernization lessons into repeatable assets. This combination is materially different from the knowledge of a general market data engineer who may know Python, Spark, AWS, or Databricks but would not normally possess LTIM Canvas context, CDM dependency knowledge, existing conversion decisions, validation history, and authored migration accelerators on day one.

6. LTIM Proprietary Platform Knowledge Applied

LTIM Area	Specialized Knowledge	How It Supports the U.S. Role
Canvas AIOps	AI-led operations, alert convergence, causal analysis, noise suppression, ITSM integration, CI/service impact, Jovi GenAI assistance, digital SOPs, session-based knowledge capture, workflow recommendations, and proactive anomaly detection.	Supports production-readiness, incident response, operational governance, monitoring alignment, release handoff, and supportability for modernized CDM pipelines.
Canvas Insights	AI/ML-led SDLC acceleration, knowledge fabric, requirement-code-defect correlation, impact analysis, test optimization, defect advisor, solution advisor, and delivery intelligence.	Supports traceability from CDM requirements to workflow dependencies, conversion decisions, impacted modules, test coverage, defect risk, and validation planning.
LTIM modernization delivery method	Legacy ETL assessment, Informatica XML analysis, framework-driven conversion, orchestration migration, reconciliation design, data quality controls, deployment governance, and release readiness.	Establishes repeatable migration execution methods that reduce risk, preserve business rules, and improve modernization consistency across multiple workflow groups.

7. Capital Group CDM Project-Specific Knowledge

The role requires project-specific knowledge that was acquired through direct execution of the Capital Group CDM modernization program. This knowledge is not limited to technology syntax; it includes understanding of how legacy workflow intent, dependency sequencing, transformation logic, validation requirements, and deployment practices must be preserved during modernization.

- Knowledge of CDM modernization objectives, workflow families, migration phases, and delivery constraints.
- Understanding of Informatica workflows, sessions, mappings, sources, targets, transformations, and dependency metadata relevant to CDM migration.
- Awareness of how legacy business rules and transformation logic must be preserved while converting to Python/PySpark and cloud-native execution models.
- Knowledge of orchestration migration from legacy scheduling concepts to Airflow/Databricks-aligned dependency handling.
- Experience with CDM-specific reconciliation, validation, audit, exception handling, and production-readiness requirements.

- Understanding of onsite/offshore delivery expectations, stakeholder coordination points, and modernization risk areas already identified during the program.

8. Project-Specific Accelerators and Validation Frameworks

The accelerators below should be described as self-authored or self-created Python engineering accelerators and as supporting evidence of advanced engineering capability. They should not be presented as the only basis for the specialized-knowledge petition, because publicly available packages can be accessed by others. The primary L-1B basis remains LTIM proprietary platform knowledge and Capital Group CDM project-specific implementation knowledge.

Accelerator	Purpose	Contribution to Proposed Role
FastRecon	Reconciliation and validation accelerator for comparing outputs across files, SQL tables, SQL queries, and heterogeneous migration outputs.	Improves source-to-target validation quality and reduces manual comparison effort between legacy Informatica outputs and modernized pipelines.
informatica-python	Python accelerator for extracting and analyzing Informatica PowerCenter XML metadata such as workflows, sessions, mappings, transformations, sources, targets, and dependencies.	Transforms legacy ETL knowledge into repeatable metadata analysis and migration assessment capability.
informatica-sparker	Informatica-to-PySpark acceleration framework supporting Spark/PySpark-aligned structures from Informatica metadata.	Accelerates conversion standardization and supports reusable PySpark engineering patterns.
Internal reusable ETL framework patterns	Configuration-driven processing, logging, audit tracking, exception handling, validation, and deployment patterns for modernized pipelines.	Improves repeatability, maintainability, governance, and operational readiness across the CDM modernization workstream.

9. Tools, Technologies, and Platforms

Category	Relevant Tools / Knowledge Areas
Legacy ETL	Informatica PowerCenter, workflows, sessions, mappings, transformations, source-target metadata, parameterization, AutoSys scheduling concepts
Modern Data Engineering	Python, PySpark, SQL, Databricks, AWS, lakehouse-aligned pipeline design, reusable ETL framework components
Orchestration and Deployment	Apache Airflow, Databricks job orchestration, CI/CD deployment concepts, release governance, environment promotion, rollback/readiness planning
Validation and Quality	FastRecon, reconciliation methods, source-to-target comparison, data quality checks, audit tables, exception tracking, validation evidence packs
Operations and Observability	Canvas AIOps concepts, Datadog/CloudWatch-aligned monitoring concepts, alert correlation, incident analysis, SOP design, production support readiness
SDLC Intelligence	Canvas Insights concepts, traceability, impact analysis, test optimization, defect linkage, knowledge-fabric-style delivery visibility

10. Expected Deliverables

- CDM workflow and dependency analysis notes for prioritized Informatica migration scope.
- Conversion guidance and reusable Python/PySpark design patterns for selected workflow groups.
- Source-to-target reconciliation approach and validation evidence templates for migrated flows.
- Operational readiness recommendations aligned to Canvas AIOps concepts, including monitoring, incident handling, and SOP readiness.
- SDLC traceability and impact-analysis guidance aligned to Canvas Insights concepts.
- Technical coordination between onsite stakeholders and offshore modernization teams.
- Architecture and design walkthrough materials for CDM modernization decision-making.
- Deployment readiness checklist, audit/control notes, failure-handling recommendations, and runbook support artifacts.
- Knowledge-transfer documentation for U.S. and offshore teams to reduce ramp-up dependency and preserve project continuity.

11. Why the U.S. Assignment Requires This Employee

The proposed U.S. assignment requires continuity of knowledge already acquired through LTIM India delivery and CDM modernization execution. A general market Senior Data Engineer may have strong Python, Spark, AWS, Databricks, or Airflow experience, but would not normally have the combined knowledge of LTIM Canvas platforms, LTIM modernization delivery practices, Capital Group CDM workflow dependencies, existing conversion decisions, validation strategy, and the project-specific accelerators authored and applied by Nikhil.

Placing Nikhil in the U.S. role supports faster onsite decision-making, reduced re-discovery of legacy workflow behavior, better preservation of transformation logic, stronger validation discipline, and smoother coordination between client stakeholders, onsite program leadership, offshore engineering teams, QA teams, and production support groups.

12. Required Qualifications for Proposed Role

- Advanced data engineering experience with enterprise ETL modernization, Python, PySpark, SQL, cloud data platforms, and orchestration design.
- Prior LTIMindtree employment and direct exposure to LTIM proprietary platforms or methods, including Canvas AIOps, Canvas Insights, and LTIM modernization practices.
- Direct Capital Group CDM modernization experience, including workflow dependency analysis, Informatica metadata assessment, transformation preservation, validation controls, and deployment readiness.
- Ability to lead technical discussions with onsite and offshore stakeholders and translate legacy implementation context into modernized cloud engineering decisions.
- Experience designing reusable conversion, reconciliation, logging, audit, and validation frameworks that improve repeatability and delivery quality.
- Ability to protect confidential client and LTIM information while producing sanitized technical documentation for project execution and immigration evidence support.

13. Comparison with General Market Role

Area	General Senior Data Engineer	Proposed Employee's Specialized Contribution
Python/PySpark capability	Can build pipelines and transformations.	Applies Python/PySpark in the context of LTIM conversion methodology, CDM workflow dependencies, reusable framework design, validation history, and deployment readiness.
Cloud data platform knowledge	May know AWS, Databricks, Airflow, and CI/CD.	Knows how those technologies are being applied in the CDM modernization context and how LTIM delivery standards should guide migration execution.
Operational readiness	May add logging and monitoring generally.	Applies LTIM Canvas AIOps-aligned thinking for incident intelligence, alert correlation, SOP readiness, monitoring, and supportability of modernized data pipelines.
SDLC traceability	May follow standard agile delivery practices.	Applies Canvas Insights concepts for requirement-code-defect correlation, impact analysis, testing impact, and modernization traceability.
Migration accelerators	May use standard tools or write one-off scripts.	Created and applied accelerators such as FastRecon, informatica-python, and informatica-sparker to standardize migration analysis, reconciliation, and conversion support.
CDM project knowledge	No inherent knowledge of the client implementation.	Possesses direct knowledge of CDM modernization objectives, dependencies, conversion decisions, validation requirements, and risk areas acquired through execution.

14. Evidence and Documentation Checklist

The following supporting materials can strengthen the proposed job description and specialized-knowledge positioning. The immigration team should decide which documents are appropriate to include in the final petition package.

- Manager support letter confirming LTIM employment history, role on Capital Group CDM, and why the U.S. assignment requires Nikhil's specialized knowledge.
- Project allocation records or HR documentation showing assignment to the CDM modernization program.
- Canvas AIOps and Canvas Insights training records, certifications, workshop attendance, internal learning screenshots, or completion evidence.
- Sanitized CDM architecture diagrams, workflow dependency summaries, or migration design documents approved for immigration use.
- Sanitized examples of Informatica XML analysis, conversion templates, reconciliation outputs, validation summaries, or migration decision records.
- Documentation or screenshots for FastRecon, informatica-python, and informatica-sparker showing authorship, usage, or technical purpose.
- PyPI/GitHub/repository evidence for public accelerators, positioned only as supporting evidence of advanced engineering capability and not as the sole L-1B basis.

- Stakeholder emails, meeting notes, design review participation, or delivery artifacts showing technical ownership and specialized contribution.
- Proposed organization chart showing reporting line, onsite/offshore collaboration, and how the U.S. role fits into CDM delivery governance.

15. Supporting Public Evidence Links

The following public links may be used as objective third-party supporting evidence for authored engineering accelerators. These links should be included only after confirming ownership/authorship evidence and counsel approval.

Reference	URL
FastRecon website	https://www.fastrecon.net/
FastRecon on PyPI	https://pypi.org/project/fastrecon/
informatica-python on PyPI	https://pypi.org/project/informatica-python/
informatica-sparker on PyPI	https://pypi.org/project/informatica-sparker/

16. Suggested Manager Attestation Language

The following paragraph may be adapted by LTIM management or immigration counsel. It should be reviewed and finalized by the authorized signatory.

Nikhil Subhash Shinde possesses specialized knowledge of LTIMindtree tools, delivery methods, and Capital Group CDM implementation practices that is materially different from the knowledge of a general Senior Data Engineer. His knowledge combines Canvas AIOps, Canvas Insights, LTIM-led Informatica-to-Python/PySpark modernization methods, CDM workflow dependency and validation context, and self-created Python accelerators such as FastRecon, informatica-python, and informatica-sparker. His proposed U.S. assignment requires this knowledge to preserve delivery continuity, reduce migration risk, accelerate validation, support onsite/offshore coordination, and advance the client's cloud modernization roadmap.

17. Final Positioning Statement

The proposed U.S. role should be positioned as a specialized-knowledge assignment tied to LTIMindtree's proprietary platforms, LTIM modernization delivery methods, and Capital Group CDM project-specific implementation experience. The value of the assignment is not merely that Nikhil can code in Python or PySpark; the value is that he already carries the combined context required to preserve legacy ETL behavior, guide conversion decisions, support validation, align production-readiness with LTIM Canvas AIOps thinking, and maintain delivery continuity for the CDM modernization program.

This document is a technical and business job-description draft for immigration review. It is not legal advice and should be finalized by LTIMindtree immigration counsel before use in any filing or formal submission.